

Spring 2021 Real Analysis II — Homework 1

FirstName LastName

ID #: 123-45-6789

1. Let (X, d) be a metric space. Show that a Cauchy sequence $\{x_k\}$ in X is convergent iff $\{x_k\}$ has a convergent subsequence.

Proof. $(\Rightarrow)$ The claim holds trivially.

$(\Leftarrow)$ Suppose a subsequence $\{x_{k_j}\}$ of $\{x_k\}$ converges to x . Then we know that, for any $\epsilon > 0$, there exist $K = K(\epsilon) \in \mathbb{N}$ and $J = J(\epsilon) \in \mathbb{N}$ such that $k_J \geq K$, $d(x_k, x_l) < \epsilon/2$ and $d(x_{k_j}, x) < \epsilon/2$ for all $k, l \geq K$ and $j \geq J$. Then

$$d(x_k, x) \leq d(x_k, x_{k_J}) + d(x_{k_J}, x) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon,$$

for all $k \geq K$. Therefore $x_k \rightarrow x$. □

2. Let (X, d) be a complete metric space and $\{x_k\}$ a given sequence. Suppose for any $\epsilon > 0$, there exists a Cauchy sequence $\{y_k\}$ such that $d(x_k, y_k) < \epsilon$ for all $k \in \mathbb{N}$. Show that $\{x_k\}$ is Cauchy.

Proof. For any ϵ , there exists a Cauchy sequence $\{y_k\}$ such that $d(x_k, y_k) < \epsilon/3$ for all $k \in \mathbb{N}$. Since $\{y_k\}$ is Cauchy, we know there exists $K = K(\epsilon) \in \mathbb{N}$, such that $d(y_{k+p}, y_k) < \epsilon/3$ for all $k \geq K$ and $p \in \mathbb{N}$. Therefore,

$$d(x_{k+p}, x_k) \leq d(x_{k+p}, y_{k+p}) + d(y_{k+p}, y_k) + d(y_k, x_k) < 3 \cdot \frac{\epsilon}{3} = \epsilon$$

for all $k \geq K$ and $p \in \mathbb{N}$, which implies that $\{x_k\}$ is Cauchy. □

3. Let (X, d) be a metric space and M a sequentially compact subset of X . Suppose the mapping $f : X \rightarrow M$ satisfies $d(f(x_1), f(x_2)) < d(x_1, x_2)$ for any distinct $x_1, x_2 \in X$. Show that f has a unique fixed point in X .

Proof. It is clear that f is continuous. Define $g : X \rightarrow \mathbb{R}$ by $g(x) := d(x, f(x))$. Then g is also continuous. Denote $\delta := \inf_{x \in M} g(x)$, then there exists a sequence $\{x_k\}$ in M such that $g(x_k) \rightarrow \delta$. Since M is sequentially compact, there exists a subsequence $\{x_{k_j}\}$ and x^* in M such that $x_{k_j} \rightarrow x^*$ as $j \rightarrow \infty$. Thus $g(x_{k_j}) \rightarrow g(x^*)$, and we know $g(x^*) = \delta = \min_{x \in M} g(x)$. If $d(x^*, f(x^*)) = g(x^*) = \delta > 0$, then we know $d(f(x^*), f(f(x^*))) < d(x^*, f(x^*)) = \delta$, i.e., $g(f(x^*)) < \delta$ which contradicts to δ being the minimal value of g on M . Hence $g(x^*) = \delta = 0$, i.e., $d(x^*, f(x^*)) = 0$ and x^* is a fixed point of f .

If $x, x' \in X$ are both fixed points of f but distinct, then $d(x, x') = d(f(x), f(x')) < d(x, x')$ which is a contradiction. Hence the fixed point of f is unique. □

4. Convergence of Newton's method. Suppose $f \in C^2([a, b])$ and $\hat{x} \in (a, b)$ such that $f(\hat{x}) = 0$ and $f'(\hat{x}) \neq 0$. Show that there exists an open neighborhood $U_{\hat{x}}$ of $\hat{x}$, such that for any $x_0 \in U_{\hat{x}}$, the sequence $\{x_k\}$ generated by Newton's method from x_0 as follows,

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}, \quad k = 1, 2, \dots,$$

is convergent, and $\lim_{k \rightarrow \infty} x_k = \hat{x}$.

Proof. WLOG assume $f'(\hat{x}) =: l > 0$. Since $f \in C^2([a, b])$, there exists $L > 0$ such that $|f''(x)| \leq L$ for all $x \in [a, b]$. Moreover, there exists $\delta_1 > 0$ such that $f'(x) > l/2$ for all $x \in (\hat{x} - \delta_1, \hat{x} + \delta_1)$, and $\delta_2 > 0$ such that $|f(x)| < l^2/(8L)$ for all $x \in (\hat{x} - \delta_2, \hat{x} + \delta_2)$. Then let $\delta < \min(\delta_1, \delta_2)$, we know $|f(x)f''(x)/(f'(x))^2| \leq 1/2$ for all $x \in (\hat{x} - \delta, \hat{x} + \delta)$ (let δ smaller to make $U_{\hat{x}} := (\hat{x} - \delta, \hat{x} + \delta) \subset (a, b)$). Note that $\hat{x}$ is the only zero of f on $U_{\hat{x}}$ due to $f'(x) > l/2$ on $U_{\hat{x}}$ and the mean value theorem. Define $T : U_{\hat{x}} \rightarrow \mathbb{R}$ by $T(x) := x - \frac{f(x)}{f'(x)}$, then

$$|T'(x)| = \left| 1 - \frac{(f'(x))^2 - f(x)f''(x)}{(f'(x))^2} \right| = \left| \frac{f(x)f''(x)}{(f'(x))^2} \right| \leq \frac{1}{2}.$$

Note that $T(\hat{x}) = \hat{x}$, i.e., $\hat{x}$ is a fixed point of T . Moreover, for any $x \in U_{\hat{x}}$, there exists, by mean value theorem, θ between x and $\hat{x}$ such that

$$|T(x) - \hat{x}| = |T(x) - T(\hat{x})| = |T'(\theta)(x - \hat{x})| \leq \frac{1}{2}|x - \hat{x}| < \frac{\delta}{2} < \delta.$$

Therefore, $T(x) \in U_{\hat{x}}$, which means $T(U_{\hat{x}}) \subset U_{\hat{x}}$, and T is contractive. By Banach contractive mapping theorem, we know T has a unique fixed point on $U_{\hat{x}}$, which is $\hat{x}$. As the Newton's method is generating $x_{k+1} = T(x_k)$, we know from the proof of the Banach contractive mapping theorem that $x_k \rightarrow \hat{x}$. $\square$

5. Let $X = \{f : [0, 1] \rightarrow \mathbb{R} : f(0) = 0, \|f\|_{\text{TV}} < \infty\}$ where

$$\|f\|_{\text{TV}} := \sup_{\Delta} \sum_{i=1}^n |f(x_i) - f(x_{i-1})|,$$

and $\Delta : 0 = x_0 < x_1 < \dots < x_n = 1$ is a partition of $[0, 1]$. Namely, X is the collection of functions who take value 0 at $x = 0$ and has finite total variation. Define $d(f, g) = \|f - g\|_{\text{TV}}$. Show that (X, d) is a complete metric space.

Proof. It is easy to show that d is nonnegative, symmetric, and satisfies triangle inequality. If $d(f, g) = 0$ then $f - g$ is constant, which implies that $f = g$ for all x since $f(0) = g(0) = 0$.

Let $\{f_k\}$ be a Cauchy sequence in X . Then for any $x \in (0, 1]$, there is

$$|f_{k+p}(x) - f_k(x)| \leq \sup_{\Delta} \sum_{i=1}^n |(f_{k+p}(x_i) - f_k(x_i)) - (f_{k+p}(x_{i-1}) - f_k(x_{i-1}))| \leq \|f_{k+p} - f_k\|_{\text{TV}} \rightarrow 0,$$

as $k \rightarrow \infty$ and for any $p \in \mathbb{N}$. This implies that $\{f_k(x)\}$ is a Cauchy sequence in $\mathbb{R}$ and hence convergent. Let $f : [0, 1] \rightarrow \mathbb{R}$ be defined by $f(x) := \lim_k f_k(x)$. Then for any $\epsilon > 0$ and any partition $\Delta : 0 = x_0 < x_1 < \dots < x_n = 1$, there exists $K = K(\epsilon) \in \mathbb{N}$, such that for any $k \geq K$, there is

$$\begin{aligned} & \sum_{i=1}^n |(f_k(x_i) - f(x_i)) - (f_k(x_{i-1}) - f(x_{i-1}))| \\ &= \lim_{p \rightarrow \infty} \sum_{i=1}^n |(f_k(x_i) - f_{k+p}(x_i)) - (f_k(x_{i-1}) - f_{k+p}(x_{i-1}))| \leq \epsilon, \end{aligned}$$

where the last inequality is because

$$\sum_{i=1}^n |(f_k(x_i) - f_{k+p}(x_i)) - (f_k(x_{i-1}) - f_{k+p}(x_{i-1}))| \leq \|f_k - f_{k+p}\| \leq \epsilon$$

for all $k \geq K$ and $p \in \mathbb{N}$ (since $\{f_k\}$ is Cauchy in X). Hence we know

$$d(f_k, f) = \|f_k - f\|_{\text{TV}} = \sup_{\Delta} \sum_{i=1}^n |(f_k(x_i) - f(x_i)) - (f_k(x_{i-1}) - f(x_{i-1}))| \leq \epsilon$$

This implies that $\|f_k - f\|_{\text{TV}} \rightarrow 0$ as $k \rightarrow \infty$, and $\|f\|_{\text{TV}} \leq \|f_K\|_{\text{TV}} + \|f_K - f\|_{\text{TV}} \leq \|f_K\|_{\text{TV}} + \epsilon < \infty$, which means $f \in X$. Therefore $f_k \rightarrow f$ in X . Hence X is complete. $\square$

6. Let (X, d) be a metric space, and E and F two compact subsets of X . Show that there exist $x \in E$ and $y \in F$ such that

$$d(x, y) = d(E, F) := \inf_{x' \in E, y' \in F} d(x', y').$$

Proof. Let $\{x_k\}$ and $\{y_k\}$ be the sequences on E and F respectively, such that $d(x_k, y_k) \rightarrow d(E, F)$. Since E and F are compact in the metric space X , we know they are sequentially compact and therefore there exist subsequences $\{x_{k_j}\}$ and $\{y_{k_j}\}$, as well as $x \in E$ and $y \in F$, such that $x_{k_j} \rightarrow x$ and $y_{k_j} \rightarrow y$. This implies $d(x_{k_j}, y_{k_j}) \rightarrow d(x, y)$. Hence $d(x, y) = d(E, F)$. $\square$

7. Consider the integral equation

$$x(t) - \lambda \int_0^1 e^{t-s} x(s) \, ds = y(t)$$

where $y \in C([0, 1])$ is given, $\lambda \in \mathbb{R}$ is a constant such that $|\lambda| < 1$. Show that the integral equation has a unique solution $x \in C([0, 1])$. [Hint: multiply e^{-t} on both sides, and show that a unique fixed point of form $e^{-t}x(t)$ exists.]

Proof. Multiplying e^{-t} on both sides yields

$$e^{-t}x(t) = e^{-t}y(t) + \lambda \int_0^1 e^{-s}x(s) \, ds.$$

As $e^{-t} > 0$ for all $t \in [0, 1]$, it suffices to show that the problem $\tilde{x}(t) = \tilde{y}(t) + \lambda \int_0^1 \tilde{x}(s) \, ds$ has a unique solution $\tilde{x}(t) = e^{-t}x(t)$ where $\tilde{y}(t) = e^{-t}y(t)$.

Define the mapping $T : C([0, 1]) \rightarrow C([0, 1])$ by

$$(Tu)(t) = \tilde{y}(t) + \lambda \int_0^1 u(s) \, ds.$$

Hence there is

$$\|Tu - Tv\| = \left| \lambda \int_0^1 (u(s) - v(s)) \, ds \right| \leq |\lambda| \|u - v\|.$$

Hence T is contractive given that $|\lambda| < 1$. By Banach contractive mapping theorem, we know T has a unique fixed point $\tilde{x}(t)$ in $C([0, 1])$, and thus a unique solution $x(t) = e^t \tilde{x}(t)$ of the equation. $\square$

8. Let M be a bounded subset of $C([a, b])$. Show that the set

$$\left\{ F(x) = \int_a^x f(t) \, dt : f \in M \right\}$$

is sequentially compact. [Hint: use Arzela-Ascoli theorem.]

Proof. Let $L > 0$ be the bound of M , that is, $\|f\| \leq L$ for all $f \in M$. Then it is clear that $\|F\| \leq L(b-a)$ for all $F \in M$, which means M is uniformly bounded, and $|F(x) - F(y)| \leq L|x - y|$ which means F is L -Lipschitz continuous for all $F \in M$ and thus are equicontinuous. By Arzela-Ascoli theorem, M is sequentially compact. $\square$

9. Let $E = \{\sin kt : k \in \mathbb{N}\}$. Show that E is not sequentially compact in $C([0, \pi])$.

Proof. Let $\epsilon_0 = 1/2$, then for any $\delta > 0$, let $k \in \mathbb{N}$ large enough such that $\frac{\pi}{2k} < \delta$, then for $x = 0$ and $x' = \frac{\pi}{2k}$, we know $|x' - x| < \delta$ but

$$\sin(kx') - \sin(kx) = \sin(kx') = \sin\left(\frac{\pi}{2}\right) = 1 > \epsilon_0.$$

Therefore E is not equicontinuous and thus not sequentially compact. $\square$

10. Let $X := \{x = (x_1, x_2, \dots) : x_i \in \mathbb{R}, \exists K \in \mathbb{N}, \text{ s.t. } x_k = 0, \forall k \geq K\}$, i.e., X is the collection of sequences that have finitely many nonzero terms. Define

$$d(x, y) = \sup_{k \in \mathbb{N}} |x_k - y_k|.$$

Show that $d(x, y)$ is a metric on X , but X is not complete under d . [Hint: consider the sequences of form $(1, \frac{1}{2}, \dots, \frac{1}{k}, 0, \dots)$ at different truncations.]

Proof. Consider the sequence $\{x^{(n)}\}$ in X (n is the indices of the terms in this sequence) where $x^{(n)} := (1, \frac{1}{2}, \dots, \frac{1}{n}, 0, \dots)$. Then for any $p \in \mathbb{N}$, there is

$$d(x^{(n+p)}, x^{(n)}) = \sup_{k \in \mathbb{N}} |x_k^{(n+p)} - x_k^{(n)}| = \frac{1}{n+1} \rightarrow 0$$

as $n \rightarrow \infty$. Hence $\{x^{(n)}\}$ is Cauchy. However, $x^{(n)} \rightarrow x^* := (1, \frac{1}{2}, \dots)$ but $x^* \notin X$. Hence X is a metric space but not complete. $\square$